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## Data Science at Scale

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| On Campus Exam                       | Closed Book                                             |
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| <b>Additional Materials:</b><br>None | <b>Permitted Materials:</b><br>No Calculators Permitted |

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**Guidelines:**

- The **duration** of this paper is 2h
  - Word Count: No Word Count
  - Write all answers in the answer booklet provided.
  - Do not open the exam paper until instructed by the Invigilator.
  - Use only black pen or pencil.
  - Mobile and electronic devices are not allowed.
  - Do not communicate with other candidates during the exam.
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**Exam Instructions:**

Answer ALL Questions

**Question 1**

- (a) *“The application performs the function that the user expected. It can tolerate the user making mistakes or using the software in unexpected ways. Its performance is good enough for the required use case, under the expected load and data. ”*

**Q:** What is being described in the previous paragraph?

**(1 mark)**

- (b) *“They are harder to anticipate. They can be present in many nodes and cause widespread system failures. ”*

**Q:** What is being described in the previous paragraph?

**(1 mark)**

- (c) *“Its cost does not scale linearly. It also offers limited fault tolerance, even though it might be simpler to maintain.”*

**Q:** What is being described in the previous paragraph?

**(1 mark)**

- (d) *“This property relates to how easy it is for the operation team to keep a system running and for new members of the team to understand it. It also relates to how easy it is for engineers to make changes and update the system. ”*

**Q:** What is being described in the previous paragraph?

**(1 mark)**

- (e) *“The structure is implicit and defined when the data is loaded by the application. Allows more efficient changes in the format of the data.”*

**Q:** What is being described in the previous paragraph?

**(1 mark)**

**(Total 5 marks)**

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### Question 2

- (a) Describe the differences between software and hardware faults in the context of system availability.

(10 marks)

- (b) When it comes to making decisions related to the design of data-intensive systems, **explain** the trade-off between **eventual consistency** and **linearizability**, and how choices made based on this trade-off can affect the availability of said system.

(10 marks)

(Total 20 marks)

### Question 3

You are the Data Science Lead for a social media platform called Y (previously known as Squeaker) in which users can post images, videos, and/or short texts to their followers. Your team is responsible for the design of the **data systems** behind the Y platform.

- (a) Describe in detail how you would design the data storage solution of Y, taking into account the following aspects of the system:

- (i) What data store model would you adopt for the user and who-follows-whom data? Justify your answer.

(5 marks)

- (ii) As a global application used by millions of users, certain decisions have to be made to ensure the scalability of the platform. Explain and justify in detail the decisions you have made regarding the distributed data architecture and consistency guarantees. Your answer should take into account all the aspects relevant to the design of a large-scale data solution.

(25 marks)

(Total 30 marks)

### Question 4

- (a) When designing a fault-tolerant, scalable system, what are the key considerations in deciding whether to use vertical or horizontal scaling?

(10 marks)

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- (b) Explain **covariate shifts** and describe an illustrative scenario in which such shift occurred.

**(10 marks)**

- (c) In the context of a leader-follower architecture, explain **consensus** and what it represents in terms of performance, consistency and availability.

**(15 marks)**

- (d) In a context of parallel computing, describe MPI explaining how it addresses the parallel programming model

**(10 marks)**

**(Total 45 marks)**